

Team Project Plan

Project Objective

The objective of this project is to design and develop a **functional digital application** in student teams. The project follows a **practical, project-based learning approach**, covering the entire lifecycle of digital product development: from idea and requirements analysis to implementation, testing, and final presentation.

Students may complete the project using **one of two technological tracks**:

1. **Full-Code Development Track**
2. **Low-Code / No-Code / AI-Code Track**

Both tracks pursue the same learning outcomes: **designing, implementing, and presenting a working digital product**.

Development Tracks

Full-Code Development Track

Students develop a **web application using traditional programming technologies**.

Example technology stack:

Backend

- Java
- Python
- .NET
- Node.js

Frontend

- React
- Angular
- Vue

Database

- PostgreSQL
- MySQL

- MongoDB

Version control

- GitHub / GitLab

Students are responsible for implementing the **application logic, database layer, and user interface**.

Low-Code / No-Code / AI-Code Track

Students build an application using **visual development platforms and AI-assisted tools**.

Possible tools include:

Low-Code / No-Code platforms:

- Bubble
- Glide
- Adalo
- Airtable
- Microsoft Power Apps

Automation tools:

- Zapier
- Make

AI-assisted development tools:

- GitHub Copilot
- Cursor
- ChatGPT
- Replit

This track emphasizes **digital product design, system integration, automation, and AI-assisted development**.

Team Organization

Projects are completed in **teams of 3–6 students**.

Each team should assign the following roles:

Project Manager

- coordinates team activities
- manages the project backlog
- organizes team meetings

Backend Developer / Logic Developer

- implements application logic
- integrates database and APIs
- develops backend functionality or workflows

Frontend Developer / UI Designer

- designs the user interface
- implements UI components

Database / Data Specialist

- designs the data model
- manages the database structure

QA & Documentation Manager

- performs testing
- prepares project documentation

In the Low-Code track, roles may also include:

- **Automation Designer**
- **AI Integration Specialist**

Assessment Criteria

Component	Points
Project repository / project environment	10
Project management plan (Jira, GitHub Projects, Trello)	10
Technical documentation	10
Team activity and commit history	10
Final presentation	10

Maximum score: **50 points**.

Additional aspects evaluated in the Low-Code track may include:

- system integrations
- automation workflows
- use of AI tools

Project Timeline (12 Weeks)

Week 1 – Project Kick-off

Objectives:

- form project teams
- select a project idea

Tasks:

- identify a user problem
- define the target users
- list initial application features

Deliverables:

- project description
- project repository created
- initial MVP feature list

Week 2 – Requirements Analysis

Objectives:

- define system functionality

Tasks:

- prepare user stories
- create a project backlog
- plan development iterations

Deliverables:

- project backlog
- defined MVP scope

Week 3 – System Architecture Design

Objectives:

- design the system architecture

Tasks:

- select technologies
- design system architecture
- define API structure or workflows

Deliverables:

- architecture diagram
- system design description

Week 4 – Technical Setup

Objectives:

- prepare the development environment

Full-Code Track:

- backend configuration
- frontend setup
- repository configuration

Low-Code Track:

- platform setup
- initial application prototype

Deliverables:

- functional project skeleton

Week 5 – Data Model Design

Objectives:

- design the application data structure

Tasks:

- define database schema or tables

- specify relationships between entities
- configure database connection

Deliverables:

- data model diagram

Week 6 – Core Logic Implementation

Objectives:

- implement basic application functionality

Full-Code Track:

- develop API endpoints
- implement CRUD operations

Low-Code Track:

- configure workflows
- implement automation processes

Deliverables:

- core application functionality

Week 7 – User Interface Development

Objectives:

- implement the user interface

Tasks:

- create application layout
- implement UI components
- connect UI to backend or data source

Deliverables:

- basic user interface

Week 8 – System Integration

Objectives:

- integrate all system components

Tasks:

- connect frontend and backend
- integrate external services
- improve user experience

Deliverables:

- first working version of the application

Week 9 – Authentication and AI Features

Objectives:

- implement advanced features

Tasks:

- user authentication
- user roles and permissions
- optional AI integration (e.g., chatbot)

Deliverables:

- extended application functionality

Week 10 – Testing

Objectives:

- ensure application quality

Tasks:

- functional testing
- integration testing
- bug fixing

Deliverables:

- stable application version

Week 11 – Documentation

Objectives:

- prepare project documentation

Tasks:

- describe system architecture
- prepare installation instructions
- document technologies used

Deliverables:

- README file
- technical documentation

Week 12 – Final Presentation

Objectives:

- present the project results

Tasks:

- prepare project presentation
- demonstrate the application
- explain architecture and decisions

Deliverables:

- final presentation
- fully functional application

Project Deliverables

Each team must submit:

Project Repository

- application code or platform configuration
- commit history showing team activity

Project Plan

Examples:

- GitHub Projects
- Jira
- Trello

Documentation

- system architecture description
- technology overview
- installation and usage instructions

Final Presentation

A **10–15 minute presentation** demonstrating the application.

Recommended Development Methodology

Teams are encouraged to use **Agile / Scrum practices**.

Typical bi-weekly sprint cycle:

1. Sprint Planning
2. Development
3. Code Review / Testing
4. Merge / Deployment
5. Sprint Demo

Skills Developed During the Project

The project develops the following competencies:

- software system design
- web application development
- digital system integration
- use of AI in software development
- IT project management
- teamwork and collaboration